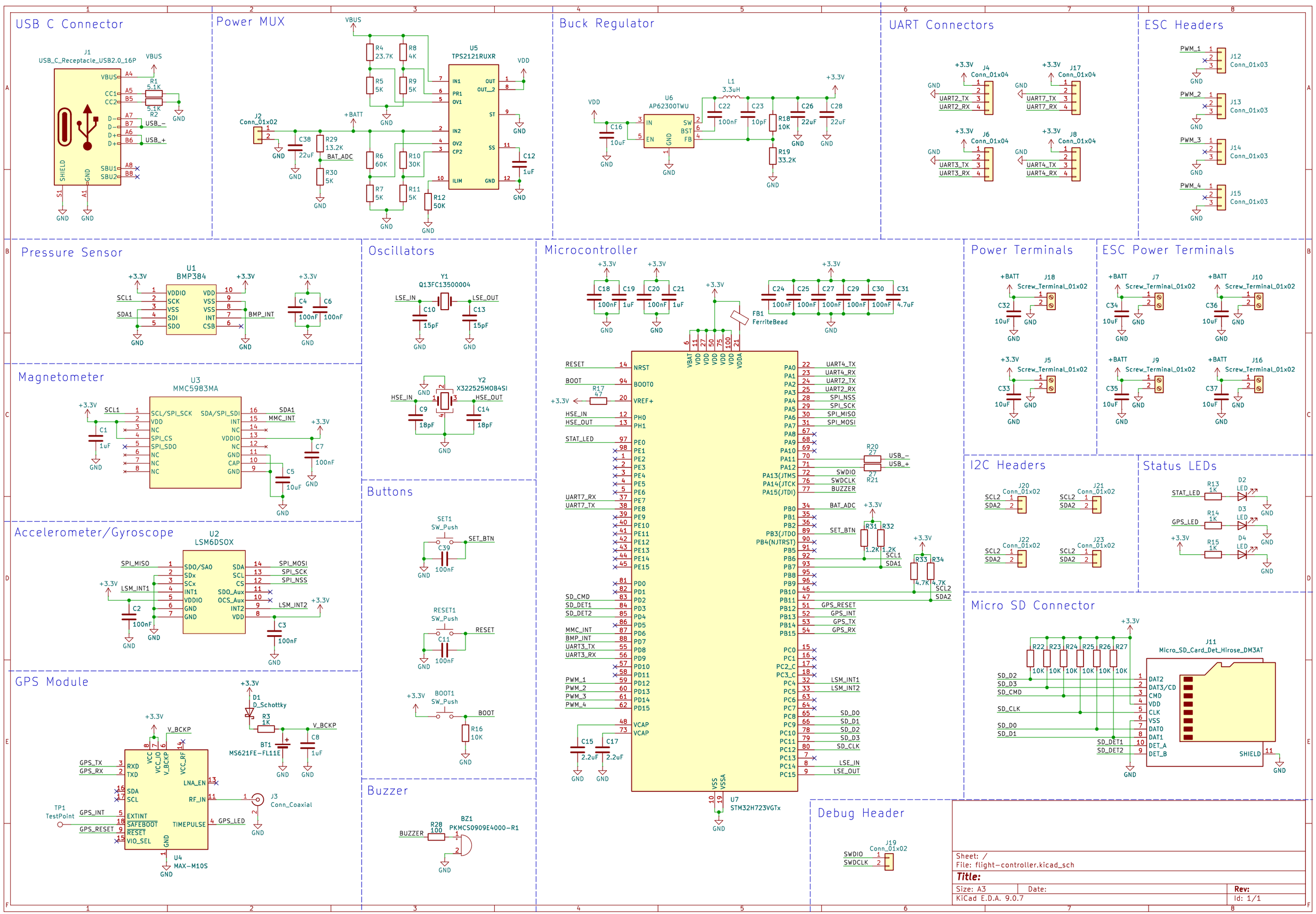
USB C Connector
J1
USB_C_Receptacle_USB2.0_16P
VBUS
R1
5.1K
R2
5.1K
CC1
CC2
D-
D+
USB_-
USB_+
SHIELD
GND
S1
A1
Power MUX
VBUS
R4
23.7K
R8
4K
R5
5K
R9
5K
U5
TPS2121RUXR
VDD
IN1
PR1
OV1
IN2
OV2
CP2
ILIM
OUT
OUT_2
ST
SS
J2
Conn_01x02
C38
22uF
R29
13.2K
BAT_ADC
R30
5K
+BATT
R6
60K
R10
30K
R7
5K
R11
5K
R12
50K
C12
1uF
Buck Regulator
U6
AP62300TWU
VDD
C16
10uF
IN
EN
SW
BST
FB
L1
3.3uH
C22
100nF
C23
10pF
R18
10K
R19
33.2K
C26
22uF
C28
22uF
+3.3V
UART Connectors
J4
Conn_01x04
UART2_TX
UART2_RX
J17
Conn_01x04
UART7_TX
UART7_RX
J6
Conn_01x04
UART3_TX
UART3_RX
J8
Conn_01x04
UART4_TX
UART4_RX
ESC Headers
PWM_1
PWM_2
PWM_3
PWM_4
J12
J13
J14
J15
Conn_01x03
Pressure Sensor
U1
BMP384
VDDIO
SCK
VSS
SDI
SDO
VDD
INT
CSB
SCL1
SDA1
BMP_INT
C4
100nF
C6
100nF
Oscillators
Y1
Q13FC13500004
LSE_IN
LSE_OUT
C10
15pF
C13
15pF
Y2
X322525MOB4SI
HSE_IN
HSE_OUT
C9
18pF
C14
18pF
Microcontroller
C18
100nF
C19
1uF
C20
100nF
C21
1uF
C24
100nF
C25
100nF
C27
100nF
C29
100nF
C30
100nF
C31
4.7uF
FB1
FerriteBead
VBAT
VDD
VDDA
RESET
BOOT
R17
47
VREF+
HSE_IN
HSE_OUT
STAT_LED
UART7_RX
UART7_TX
SD_CMD
SD_DET1
SD_DET2
MMC_INT
BMP_INT
UART3_TX
UART3_RX
PWM_1
PWM_2
PWM_3
PWM_4
NRST
BOOT0
PH0
PH1
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
VCAP
C15
2.2uF
C17
2.2uF
VSS
VSSA
U7
STM32H723VGTx
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS
PA14(JTCK
PA15(JTDI)
UART4_TX
UART4_RX
UART2_TX
UART2_RX
SPI_NSS
SPI_SCK
SPI_MISO
SPI_MOSI
R20
27
R21
27
USB_-
USB_+
SWDIO
SWDCLK
BUZZER
PB0
PB1
PB2
PB3(JTDO
PB4(NJTRST)
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
BAT_ADC
SET_BTN
R31
1.2K
R32
1.2K
SCL1
SDA1
R33
4.7K
R34
4.7K
SCL2
SDA2
GPS_RESET
GPS_INT
GPS_TX
GPS_RX
LSM_INT1
LSM_INT2
PC0
PC1
PC2_C
PC3_C
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
SD_D0
SD_D1
SD_D2
SD_D3
SD_CLK
LSE_IN
LSE_OUT
Power Terminals
J18
Screw_Terminal_01x02
C32
10uF
J5
Screw_Terminal_01x02
C33
10uF
ESC Power Terminals
J7
Screw_Terminal_01x02
C34
10uF
J10
Screw_Terminal_01x02
C36
10uF
J9
Screw_Terminal_01x02
C35
10uF
J16
Screw_Terminal_01x02
C37
10uF
Magnetometer
U3
MMC5983MA
SCL/SPI_SCK
VDD
NC
SPI_CS
SPI_SDO
SDA/SPI_SDI
INT
VDDIO
GND
CAP
C1
1uF
C7
100nF
C5
10uF
Accelerometer/Gyroscope
U2
LSM6DSOX
SDO/SA0
SDx
SCx
INT1
VDDIO
GND
SDA
SCL
CS
SDO_Aux
OCS_Aux
INT2
VDD
C2
100nF
C3
100nF
Buttons
SET1
SW_Push
SET_BTN
C39
100nF
RESET1
SW_Push
RESET
C11
100nF
BOOT1
SW_Push
BOOT
R16
10K
GPS Module
D1
D_Schottky
R3
1K
V_BCKP
BT1
MS621FE-FL11E
C8
1uF
RXD
TXD
SDA
SCL
EXTINT
SAFEBOOT
RESET
VIO_SEL
VCC
VCC_IO
V_BCKP
VCC_RF
LNA_EN
RF_IN
TIMEPULSE
GPS_LED
TP1
TestPoint
J3
Conn_Coaxial
U4
MAX-M10S
Buzzer
BZ1
PKMCS0909E4000-R1
R28
100
BUZZER
I2C Headers
J20
Conn_01x02
J21
Conn_01x02
J22
Conn_01x02
J23
Conn_01x02
SCL2
SDA2
Status LEDs
STAT_LED
GPS_LED
R13
1K
R14
1K
R15
1K
D2
LED
D3
LED
D4
LED
Micro SD Connector
J11
Micro_SD_Card_Det_Hirose_DM3AT
R22
R23
R24
R25
R26
R27
10K
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
SD_DET1
SD_DET2
Debug Header
J19
Conn_01x02
SWDIO
SWDCLK
Sheet: /
File: flight-controller.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 1/1